Lab 6 – UVM Testbench

Existing Testbench Structure

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Block	Lines	Role	UVM equivalent
Parameters & signals	1–27	Declare DUT ports directly	`mem_ctrl_if`
DUT instantiation	30–34	Connect TB to DUT	(unchanged)
Transaction type	36–50	Combined M0+M1 stimulus packet	`ctrl_transaction`
Mailboxes & flags	52–65	Inter-process communication	`ctrl_sequencer` + objections
Clock & Reset	67–77	Timing infrastructure	Top-level module
Generator (directed)	89–119	Deterministic M0/M1 stimulus	`ctrl_directed_seq`
Generator (contention)	122–137	Simultaneous M0+M1 stimulus	`ctrl_contention_seq`
Generator (random)	140–156	Randomised mixed stimulus	`ctrl_random_seq`
Driver	172–233	Drive DUT pins from mailbox	`ctrl_driver`
Checker	236–284	Golden model + result comparison	`ctrl_scoreboard`
Monitor	289–349	Sample DUT signals, build events	`ctrl_monitor`

Main control	354–397	Launch tasks,
		wait, `$finish`
			`ctrl_directed_test`
			`ctrl_contention_test`
			`ctrl_random_test`
			`ctrl_full_test`

Table 1: Structure of `tb_simple_mem_ctrl.sv` and its UVM mapping.

UVM Testbench Diagram

```
uvm_test_top (ctrl_directed_test | ctrl_contention_test |
              ctrl_random_test   | ctrl_full_test)
|
+-- ctrl_env
    |
    +-- ctrl_agent
    |   |
    |   +-- ctrl_sequencer <-------- ctrl_directed_seq
    |   |          |                  ctrl_contention_seq
    |   |    TLM handshake            ctrl_random_seq
    |   |          |
    |   +-- ctrl_driver ----drive---> mem_ctrl_if ---> DUT (simple_mem_ctrl)
    |   |                                                         |
    |   +-- ctrl_monitor <--sample-- mem_ctrl_if <-------------+
    |         |
    |    analysis_port
    |         |
    +-- ctrl_scoreboard
```